

HeartHealth AI

Practical digital cardiology education platform with AI-assisted personalization



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Model: Free risk assessment + paid subscription education + AI coach

Build Stack: Azure + Python + latest OpenAI models + Stripe

Initial Scope: 12-week MVP, evenings/weekends, 2-person team

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What We'd Be Building

A consumer-facing website where people can understand their heart health better through two integrated products:

Product 1: Free Heart Health Risk Assessment (The Front Door)

A straightforward web tool where a visitor answers a short questionnaire — age, sex, blood pressure, cholesterol numbers, smoking status, diabetes status, whether they're on BP meds — and gets back a personalized heart health report.

Under the hood, it calculates real clinical risk scores:

- **AHA/ACC ASCVD 10-Year Risk Score** (Pooled Cohort Equations) — the same calculator cardiologists use in practice to estimate 10-year risk of atherosclerotic cardiovascular events
- **Framingham Risk Score** — established coronary heart disease risk prediction
- **CHA₂DS₂-VASc Score** — stroke risk stratification for users with AFib

These are published, public-domain formulas. The platform implements them programmatically and then uses AI (OpenAI's latest models via Azure) to translate the raw numbers into a plain-English educational summary: what the score means, which risk factors are contributing the most, and general lifestyle context — all framed as education, not medical advice.

What the user actually sees:

1. A clean multi-step questionnaire (takes ~3-5 minutes)
2. A results dashboard showing their risk scores with visual indicators
3. An AI-generated narrative summary: "Based on your inputs, your estimated 10-year ASCVD risk is 8.2%, which falls in the borderline range. Your primary contributing factors are elevated LDL cholesterol and systolic blood pressure. Here's what these numbers mean..."
4. Links to deeper educational content on each risk factor
5. A prompt to create a free account to save and track scores over time

Why it matters as the free tier: It provides genuine value (people get real clinical risk scores that their doctor would calculate), builds trust through Brian's credentials, captures email addresses for the marketing funnel, and naturally leads users to want more — "okay, my cholesterol is high, now what?" — which is where Product 2 comes in.

Product 2: Paid Subscription — Content Library + AI Health Coach

For \$14.99–24.99/month, subscribers get access to:

A. Brian's Educational Content Library

Video lessons and written guides on the cardiac topics patients most commonly search for — authored and presented by Brian with his credentials displayed. Initial content priorities:

TOPIC	WHY THIS TOPIC	CONTENT FORMAT
Understanding Your Nuclear Stress Test Results	Brian is board-certified in Nuclear Cardiology. ~6,000 monthly Google searches, very few physician-authored consumer resources exist for this topic	10-min video + written guide
Understanding Your Echocardiogram Results	Brian is board-certified in Adult Echocardiography. ~8,000 monthly searches, same gap in quality consumer content	10-min video + written guide
Living With AFib: What You Need to Know	~6 million Americans have AFib. 60,000+ monthly searches. Patients are actively looking for trustworthy guidance	Video series (3–4 parts) + guide
Cholesterol & Statins: The Evidence	One of the most controversial topics in cardiology — high engagement. 40,000+ monthly searches	Video + evidence-based article
Managing Hypertension	Most common cardiac condition. Lifestyle modifications, medication education, BP monitoring guidance	Video + guide
When to See a Cardiologist	~8,000 monthly searches. Natural decision-support content that builds trust in Brian's expertise	Written guide

B. AI Health Coach (Chat)

An AI chatbot that answers subscriber questions — but grounded exclusively in Brian's curated content using a technique called Retrieval-Augmented Generation (RAG). Here's how it works:

1. Brian's videos (transcribed) and articles are broken into chunks and stored in a searchable vector database

2. When a user asks a question (e.g., "My echo showed mild mitral regurgitation, should I be worried?"), the system retrieves the most relevant chunks from Brian's content
3. The AI generates a response citing Brian's actual explanations — not generic internet content
4. Every response includes a citation: "Based on Dr. Chen's guide to understanding echocardiogram results..."
5. Safety guardrails: the AI explicitly declines to give specific medical advice and directs users to consult their physician for personal medical decisions

This is meaningfully different from asking ChatGPT a health question. The responses are grounded in content a board-certified cardiologist has reviewed, cite their source, and don't hallucinate medical claims.

C. One-Time Premium Reports (\$29-49)

For users who don't want a subscription but want a detailed one-time assessment: a comprehensive AI-generated heart health report with Dr. Chen's educational context, covering all calculated risk scores, risk factor breakdown, and relevant educational summaries. This serves as a middle-tier monetization option.

Why This Could Work — The Core Advantages

Brian's Credentials Are the Product

The digital health space is full of anonymous apps and generic content. The differentiator here is a real, practicing, board-certified cardiologist with three board certifications:

- **General Cardiology** — broad credibility across all cardiac topics
- **Nuclear Cardiology** — specialized knowledge that almost nobody is translating into consumer content
- **Adult Echocardiography** — same gap: patients get echo results and have nowhere trustworthy to go for explanations

Google's search algorithm explicitly prioritizes health content from credentialed medical professionals (they call it E-E-A-T: Experience, Expertise, Authoritativeness, Trustworthiness). Under their "Your Money or Your Life" guidelines, health pages authored by a board-certified specialist with verifiable credentials will rank higher than generic content. This is a structural SEO advantage.

The Market Context

- The global digital health market is valued at approximately **\$199 billion in 2025**, projected to reach **\$574 billion by 2030** (MarketsandMarkets)
- Consumer cardiac health is one of the fastest-growing segments, driven by wearable devices (Apple Watch ECG, Withings BP monitors) creating a large population of health-aware consumers who want to understand their data
- There's a notable gap in the market: clinical-grade tools like HeartFlow (\$1,100/study, B2B only) and Clearly (B2B only) serve physicians, while consumer tools are either hardware-dependent (KardiaMobile, \$79 + \$9.99/mo) or generic wellness apps. Nobody is offering a cardiologist-branded, AI-enhanced educational platform for consumers at an accessible price point

Regulatory Position: Wellness App, Not Medical Device

This is a critical distinction. The FDA classifies software that diagnoses, treats, or prevents disease as a "Software as a Medical Device" (SaMD), which requires formal clearance (510(k) or higher). However, software that:

- Uses published risk calculators with transparent methodology
- Provides general health education

- Does not claim to diagnose or treat specific conditions
- Includes appropriate disclaimers ("for educational purposes, consult your physician")

...falls under **FDA enforcement discretion** for general wellness products, and may also qualify for the Clinical Decision Support exemption under the 21st Century Cures Act (Section 3060). This means no FDA clearance process is needed — as long as we design it correctly from the start.

We would still want a health tech attorney (\$500–1,500 one-time) to review our approach and draft proper disclaimers and Terms of Service before launch.

How the Numbers Work

Infrastructure Costs (Monthly, at MVP Scale)

SERVICE	PURPOSE	ESTIMATED MONTHLY COST	REASONING
Azure App Service (B1 tier)	Hosts the Python/FastAPI backend API	\$13-55	B1 (\$13) handles low traffic; P1v3 (\$55) if we need more compute. Start small, scale up.
Azure Static Web Apps	Hosts the Next.js frontend	\$0-9	Free tier includes custom domain and SSL. Standard tier (\$9) adds more bandwidth.
Azure OpenAI Service (GPT-5 mini)	Powers the AI report generation and chatbot	\$20-100	GPT-5 mini costs \$0.25/1M input tokens, \$2.00/1M output tokens. A typical risk report conversation uses ~2K input + 1K output tokens = ~\$0.003/report. At 1,000 reports/month = ~\$3. At 10,000 = ~\$30. Subscriptions with ongoing chat add more. Budgeting \$20-100 to cover growth.
Azure AI Search (Basic tier)	Vector store for RAG chatbot — stores and retrieves Brian's content embeddings	\$75	Basic tier supports up to 5 indexes and 2GB storage. More than enough for initial content library. This is the priciest fixed cost at MVP.
Azure Cosmos DB (serverless) or PostgreSQL Flexible (Burstable B1ms)	User accounts, saved risk scores, subscription status	\$25-50	Serverless Cosmos: ~\$0.25 per 1M RUs, minimal at low scale. PostgreSQL Burstable B1ms: ~\$25/mo. Either works.
Azure Blob Storage	Stores video content and static assets	\$5-10	Hot tier: \$0.02/GB. 100GB of video content = \$2. Plus CDN egress.

SERVICE	PURPOSE	ESTIMATED MONTHLY COST	REASONING
Azure AD B2C	User authentication (signup, login, password reset)	\$0	Free for the first 50,000 monthly active users. We won't hit that for a long time.
Stripe	Payment processing	2.9% + \$0.30/tx	No monthly fee. On a \$24.99 subscription: Stripe takes ~\$1.02, we net ~\$23.97.
ConvertKit or SendGrid	Email marketing and transactional email	\$0-30	Free tiers cover initial needs (up to 1,000 subscribers on ConvertKit, 100 emails/day on SendGrid free).
Total estimated monthly		\$140-330/month	Conservative range. Actual cost depends on traffic. Many services have pay-as-you-go pricing that starts near zero.

Why these estimates are reasonable: Azure's pay-as-you-go model means costs scale with usage. At launch with tens or hundreds of users, actual costs will be at the low end. The AI Search Basic tier (\$75) is the one fixed cost that doesn't scale down — but it's necessary for the RAG chatbot to work well.

Initial Investment (\$1,000-3,000)

ITEM	ESTIMATED COST	REASONING
Health tech attorney consultation	\$500-1,500	1-2 hour consultation to confirm regulatory classification, review disclaimers/ToS, advise on California business entity structure (corporate practice of medicine rules). Rates vary; health tech attorneys in CA typically charge \$300-500/hr.
Domain registration	\$12-50/year	Standard .com domain. Premium domains cost more but aren't necessary.
Brand identity (logo, basic design)	\$0-200	Can start with Canva or similar. Professional logo later if traction warrants it.

ITEM	ESTIMATED COST	REASONING
First 3 months of Azure infrastructure	\$420–990	Based on the \$140–330/month estimate above.
Google Ads validation test	\$100–200	2–4 weeks of ads at \$5–10/day targeting cardiac health keywords. Purpose: measure demand (email signup rate) before building the full product.
Total	\$1,030–2,940	

Why \$1K–3K is sufficient: My 15+ years of development experience eliminates the need for hired developers. Brian's content creation replaces hiring a medical writer. Azure's pay-as-you-go pricing and free tiers minimize upfront infrastructure costs. The main non-optional expense is the attorney.

Revenue Projections and Reasoning

Assumptions behind the numbers:

METRIC	ASSUMPTION	REASONING
Monthly site visitors (month 6)	2,000–5,000	Based on targeting SEO keywords with 6,000–60,000 monthly searches. Even capturing 1–3% of search traffic for low-competition terms (nuclear stress test results, echo results) yields this range. Paid ads supplement organic traffic early on.
Free risk assessment completion rate	30–50% of visitors	Health risk calculators typically see high engagement when they're free and fast. The AHA's own calculator gets millions of uses.
Email capture rate	40–60% of assessment completers	Users must create an account to save results. Offering ongoing tracking is a strong incentive.
Free → paid conversion rate	3–5%	Industry benchmark for freemium health apps is 2–5%. Our advantage: the free tool naturally creates "what do I do about this?" questions that the paid tier answers.

METRIC	ASSUMPTION	REASONING
Average subscription price	\$19.99/month	Midpoint of \$14.99–24.99 range. Competitive with KardiaCare (\$9.99/mo for ECG-only) while offering broader educational value.
Monthly churn rate	8–12%	Health subscription apps typically see 8–15% monthly churn. Educational content with ongoing value (new content, AI chat) should be at the lower end.
One-time report purchases	5–10% of non-subscribing assessment completers	Lower-commitment option for users not ready for a subscription.

Projected Revenue Timeline:

TIMEFRAME	SUBSCRIBERS (EST.)	ONE-TIME REPORTS/MO	MONTHLY REVENUE	REASONING
Month 1–2	0	0	\$0	Building MVP, no product live yet.
Month 3 (launch)	10–30	5–15	\$200–850	Initial launch to waitlist + early organic traffic. Small numbers, proving the funnel works.
Month 6	50–150	20–50	\$1,600–5,000	SEO starting to bring in organic traffic. Content library growing. Word of mouth from early users.
Month 9	150–400	40–80	\$4,400–11,600	Compound effect of monthly content, growing SEO authority, email nurture converting backlog.

TIMEFRAME	SUBSCRIBERS (EST.)	ONE-TIME REPORTS/ MO	MONTHLY REVENUE	REASONING
Month 12	300-800	60-120	\$8,400-21,800	Established SEO rankings, YouTube channel supplementing, email list of 5,000-10,000.

Why these projections are conservative: - They assume organic SEO as the primary growth driver, which takes 3-6 months to gain traction - They don't account for potential viral social media content (Brian's short-form videos could accelerate growth unpredictably) - They assume no B2B or employer wellness revenue, which could be explored later - Churn is modeled at the higher end (10%) to be conservative - Comparable models: Dr. Peter Attia's content membership (estimated \$5M+ ARR) and Examine.com (millions in ARR) demonstrate the ceiling for physician-led health content subscriptions, though those are much more established brands

Break-even analysis: At ~\$250/month infrastructure cost, we break even at roughly 13 subscribers at \$19.99/month. That's achievable within the first month of launch.

What Each of Us Brings / Time Commitment

Brian's Role: Clinical Authority & Content Creator

Estimated time: 3-5 hours/week

TASK	FREQUENCY	TIME	WHAT IT INVOLVES
Content review and approval	Weekly	1-2 hrs	Review AI-generated educational summaries, system prompts, and content templates for clinical accuracy. I'll draft; Brian reviews.
Video recording	2-3x/month	1-2 hrs/session	Record educational videos on priority topics. Can be filmed at home or office with a good webcam — production value matters less than the content and credentials.
Article review	2-3x/month	30-60 min	Review AI-drafted articles (I'll use AI to create first drafts based on medical guidelines, Brian edits for accuracy and adds clinical nuance).
AI chatbot spot-checking	Monthly	30-60 min	Review a sample of AI chatbot responses to ensure quality and accuracy. Flag any issues for me to address in the system prompts.

What Brian does NOT need to do: Write code, manage infrastructure, handle billing, do marketing, interact with individual users, or provide any form of personalized medical advice through the platform.

My Role: Everything Technical + Product/Marketing

Estimated time: 10-15 hours/week (evenings/weekends)

TASK	PHASE	TIME
Backend development (Python/FastAPI, risk calculators, AI integration)	Phase 1	6-8 weeks
Frontend development (Next.js, responsive design)	Phase 1	4-6 weeks (parallel with backend)
Stripe integration and subscription management	Phase 1	1-2 weeks

TASK	PHASE	TIME
RAG chatbot pipeline (Azure AI Search + OpenAI)	Phase 2	2-3 weeks
SEO optimization, schema markup, content publishing	Ongoing	2-3 hrs/week
Infrastructure management, monitoring, updates	Ongoing	1-2 hrs/week
Email marketing setup and automation	Phase 1-2	1 week initial, then maintenance

Timeline

WEEKS	MILESTONE	KEY DELIVERABLE
1-4	Validation & Legal	Landing page live, ad test running, attorney consulted, partnership terms agreed, domain secured
5-8	MVP Development	Risk assessment questionnaire, score calculation engine, AI report generation, basic frontend, user accounts
9-12	Launch-Ready	Stripe payments integrated, premium report purchase flow, subscription management, disclaimers and legal copy finalized, Brian's first 3-5 content pieces live
10-16	Content + AI Coach (overlapping)	RAG chatbot live, Brian's content library growing (target: 10+ pieces), blog posts publishing for SEO
16-24	Growth Phase	YouTube channel, social media presence, email list at 500-1,000, first meaningful revenue month, iterating based on user feedback

Why 12 weeks to launch (not faster): Even with my experience, building a health product at evenings/weekends pace (10-15 hrs/week) takes time. The risk calculators themselves are straightforward, but the AI integration, payment flows, authentication, content management system, and frontend polish all add up. Doing it right — especially the disclaimers, the AI safety guardrails, and the attorney review — is more important than doing it fast. I'd rather launch a solid product in week 12 than a fragile one in week 6.

Why not slower: The core technical work (API + frontend + Stripe) is well within my wheelhouse. I've built similar stacks professionally. The risk calculators are published formulas. Azure OpenAI and Stripe both have excellent documentation and SDKs. There's no R&D uncertainty in the tech stack — it's execution.

Risks and How We'd Handle Them

RISK	LIKELIHOOD	IMPACT	MITIGATION
FDA reclassifies the tool as SaMD	Low	High	Design strictly as education/wellness from day one. Attorney confirms classification. No diagnostic claims. Transparent methodology. Robust disclaimers.
Low organic traffic / slow SEO	Medium	Medium	Supplement with small Google Ads budget (\$100–200/mo). Target long-tail, low-competition keywords first (nuclear stress test results, echo results). SEO is a 3–6 month game by nature.
AI generates inaccurate medical information	Medium	High	RAG architecture constrains AI to Brian's reviewed content. System prompts enforce educational framing and "consult your doctor" guardrails. Brian spot-checks monthly. We can add automated quality checks over time.
Low conversion from free to paid	Medium	Medium	Iterate on pricing, content value proposition, and email sequences. The free→paid funnel is the most testable part of the business — we can experiment quickly.
Brian's time becomes a bottleneck	Low-Medium	Medium	Frontload content creation (batch recording sessions). Use AI to draft content that Brian reviews/edits rather than writing from scratch. The platform is designed to scale without per-user physician time.
Competitor enters the space	Low	Low-Medium	Brian's credentials and personal brand are not replicable. A competitor would need their own board-certified cardiologist willing to be the face of the product. First-mover advantage in the nuclear cardiology and echo content niches compounds over time.

RISK	LIKELIHOOD	IMPACT	MITIGATION
HIPAA concerns	Low	High	Minimize collection of PHI. Use Azure HIPAA-eligible services with BAA. The wellness/education framing reduces HIPAA applicability, but we'd follow best practices (encryption, access controls) regardless.

What Success Looks Like

6-month check-in targets (realistic, not aspirational):

- Platform live with free risk assessment + paid subscription
- 50–150 paying subscribers
- 5–10 pieces of Brian's content published
- Email list of 1,000–3,000
- Breaking even on infrastructure costs
- Ranking on page 1 of Google for at least 2–3 target keywords
- Clear data on what content resonates and what the conversion funnel looks like

12-month targets:

- 300–800 subscribers, \$8K–22K/month recurring revenue
- 10,000+ email list
- Brian's content library: 20+ videos/articles
- YouTube channel with consistent uploads
- Data to decide whether to pursue Phase 3 (remote interpretation services) or double down on content

Decision point at 6 months: If after 6 months the metrics don't show a credible path to the 12-month targets, we'd have spent roughly \$2,500–5,000 total and can make an informed decision about whether to pivot, scale down, or wind it down. The sunk cost is manageable.

Ownership & Business Structure

This needs to be a conversation between us, but here are the standard frameworks for a two-person side project:

- **50/50 partnership** — simplest, appropriate if we view contributions as equally valuable (clinical expertise/content vs. technical/operational)
- **Proportional to contribution** — weighted by time, capital, or IP value; more complex to track
- **Vesting schedule** — equity vests over 2–4 years with a 1-year cliff, protecting both parties if one steps back early

The partnership agreement should also cover: decision-making process, what happens if one person wants out, IP ownership, and how additional revenue streams (e.g., if Brian's content leads to speaking opportunities) are handled.

The attorney consult would cover the right business entity for California (likely an LLC, but the corporate practice of medicine doctrine in CA requires that the medical content liability sits appropriately — the attorney will advise).

Next Step

If this direction makes sense, the immediate action items are:

1. **Agree on partnership terms** (even a handshake framework to start)
2. **Pick a brand name and secure the domain**
3. **I build the landing page and start the validation ad test** (1-2 weeks)
4. **Schedule the attorney consult** (can happen in parallel)
5. **Brian identifies his first 3 content topics** and we block time for a recording session

The landing page validation costs ~\$100-200 and takes 2 weeks. If the signup rate is encouraging, we proceed with the full build. If not, we've spent very little and learned something useful.